

Yu Sheng Aow

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EDUCATION

Boston University

Bachelor of Arts in Computer Science; Dean's List (Fall 2025, Spring 2026)

Boston, MA

June 2027

PROFESSIONAL EXPERIENCE

Quansight PBC

Open-Source Engineer Intern

Jul 2026 – Sep 2026

Remote

- **MLIR Backend Development:**

- * **Feature Design:** Devised a lowering strategy to translate a sparse array programming framework into MLIR. Prototyped MLIR kernels to validate mapping of compiler loops and operations to MLIR dialects.
- * **Feature Implementation:** Built an MLIR code-generation pipeline in Python with buffer management, memory ownership across lowering stages, and support for parallel computation through MLIR's GPU dialects. Explored the Linalg dialect for parallelism, vectorization, and kernel optimization.
- * **Feature Delivery:** Presented code optimization and benchmarking results in a company technical presentation, demonstrating a 2× speedup on the SDDMM kernel over SciPy backends. Authored a technical blog providing an in-depth explanation of the MLIR lowering pipeline and analysis of benchmark results.

RELEVANT EXPERIENCE

College of Computing at Georgia Tech

Undergraduate Research Assistant

Jan 2026 – Present

Boston, MA

- **MLIR GPU Backend Development:** Developed a GPU backend for a sparse array programming framework, mapping compiler loops to the MLIR GPU dialect and introducing buffer management for end-to-end compilation of dense and sparse GPU kernels, extending the framework from CPU execution to GPU architectures.
- **NVIDIA GPU Optimization:** Extended the GPU backend with the NVGPU dialect and Tensor Core support to accelerate dense computational loops and improve execution efficiency.

COMMIT Compiler Group, MIT CSAIL

Undergraduate Research Assistant

Jul 2025 – Jan 2026

Boston, MA

- **STC Kernel Execution:** Built a cost-based logical optimizer leveraging greedy search to generate synchronized execution plans for sparse tensor contractions, optimizing contraction order and kernel execution to deliver stable plans to avoid timeouts on benchmarks where modern query engines such as DuckDB and Umbra failed.
- **Pessimistic Cardinality Estimator:** Constructed a pessimistic cardinality estimator informed by real-world sparse tensor distributions, combining chain-bound estimation with breadth-first search to improve cardinality estimates and support more efficient relational query planning across complex sparse tensor contractions.
- **Tensor Interface Integration:** Added eager and lazy tensor interfaces for a sparse array library, extending functionality while ensuring compliance with the Python Array API standard.

BU Image and Video Computing Group (IVC)

Undergraduate Research Assistant

Jan 2026 – Jun 2026

Boston, MA

- **Distributed LoRA Fine-Tuning:** Trained 8B-parameter vision-language models through a Modal-based fine-tuning pipeline. Configured a CUDA GPU environment with video-processing libraries and orchestrated multi-GPU training, checkpointing, and cloud storage for reliable large-scale model development.
- **Multi-Model Evaluation Framework:** Formulated a benchmarking pipeline for vision-language models, executing batched inference across architectures, computing standardized metrics, and aggregating reproducible reports to reduce manual evaluation overhead and support consistent model comparisons.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, Assembly, Julia, OCaml

Frameworks & Libraries: PyTorch, DeepSpeed, NumPy, SciPy, pandas, Matplotlib, SuiteSparse

Compilers & Systems: MLIR, LLVM, CUDA, Linux, Git, Modal